Module 9 : Completely randomized designs with a single factor with more than two levels

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Design of Experiments - Stat 140

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Experiments with one factor with more than two levels

- Analysis of Variance (ANOVA).
- Experiment with different doses to infer dose-response.
- Can be useful for the analysis of observational studies.

Assignment mechanism

- Consider a population of N units, each of which can be exposed to any one of J treatments.
- There are $\frac{N!}{N_1!...N_J!}$ possible ways in which N experimental units can be randomly allocated to the J treatments.
- Assignment mechanism :

$$P(\mathbf{W}|\mathbf{X},\mathbf{Y(1)},...,\mathbf{Y(J)}) = \begin{cases} \frac{1}{\frac{N!}{N_1!...N_J!}} & \text{if } \sum_{i=1}^{N} \mathbf{1}(W_i = j) = N_j \in W^+ \\ 0 & \text{otherwise.} \end{cases}$$

Science Table

- Let $Y_i(W_i = j)$ denote the the ith unit potential outcome under treatment j.

Unit i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	$Y_1(W_1 = 1)$	$Y_1(W_1 = 2)$	...	$Y_1(W_1 = J)$
2	$Y_2(W_2 = 1)$	$Y_2(W_2 = 2)$	...	$Y_2(W_2 = J)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(W_N = 1)$	$Y_N(W_N = 2)$	...	$Y_N(W_N = J)$

- Causal estimands

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- Let $Y_i(W_i = j)$ denote the the ith unit potential outcome under treatment j.

Unit i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	$Y_1(W_1 = 1)$	$Y_1(W_1 = 2)$	...	$Y_1(W_1 = J)$
2	$Y_2(W_2 = 1)$	$Y_2(W_2 = 2)$	...	$Y_2(W_2 = J)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(W_N = 1)$	$Y_N(W_N = 2)$	...	$Y_N(W_N = J)$

- Causal estimands
  - Unit-level causal effects of treatment j versus j' :

$$\tau_i(j, j') = Y_i(W_i = j) - Y_i(W_i = j')$$

Science Table

- Let $Y_i(W_i = j)$ denote the the ith unit potential outcome under treatment j.

Unit i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	$Y_1(W_1 = 1)$	$Y_1(W_1 = 2)$	...	$Y_1(W_1 = J)$
2	$Y_2(W_2 = 1)$	$Y_2(W_2 = 2)$	...	$Y_2(W_2 = J)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(W_N = 1)$	$Y_N(W_N = 2)$	...	$Y_N(W_N = J)$

- Causal estimands
  - Average causal effect of treatment j versus j' :

$$\tau(j, j') = \frac{1}{N} \sum_{i=1}^{N} \tau_i(j, j') = \overline{Y}(j) - \overline{Y}(j')$$

Science Table

- Let $Y_i(W_i = j)$ denote the the ith unit potential outcome under treatment j.

Unit i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	$Y_1(W_1 = 1)$	$Y_1(W_1 = 2)$	...	$Y_1(W_1 = J)$
2	$Y_2(W_2 = 1)$	$Y_2(W_2 = 2)$	...	$Y_2(W_2 = J)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(W_N = 1)$	$Y_N(W_N = 2)$	...	$Y_N(W_N = J)$

- Causal estimands

- Note that, for each unit i, only one of the J potential outcomes can be observed.

Orthogonal decomposition of the potential outcomes

- The potential outcome $Y_i(W_i = j)$ can be written as :

$$Y_i(W_i = j) = \overline{Y} + [\overline{Y}_j - \overline{Y}] + [Y_i(W_i = j) - \overline{Y}_j]$$

where

  - $\overline{Y} = \frac{1}{J*N} \sum_{j=1}^{J} \sum_{i=1}^{N} Y_i(W_i = j)$
  - $\overline{Y}_j = \frac{1}{N} \sum_{i=1}^{N} Y_i(W_i = j)$

Orthogonal decomposition of the potential outcomes

- A measure of overall variability among the N*J potential outcomes is the Total Sum of Squares ($SSTotal_{PO}$) :

$$
\begin{aligned}
SSTotal_{PO} &= \sum_{i=1}^{N} \sum_{j=1}^{J} [Y_i(W_i = j) - \overline{Y}]^2 \\
&= \sum_{i=1}^{N} \sum_{j=1}^{J} [(\overline{Y}_j - \overline{Y}) + (Y_i(W_i = j) - \overline{Y}_j)]^2 \\
&= N \sum_{j=1}^{J} [\overline{Y}_j - \overline{Y})]^2 + \sum_{i=1}^{N} \sum_{j=1}^{J} [Y_i(W_i = j) - \overline{Y}_j]^2 \\
&= SSTreatment_{PO} + SSResidual_{PO}
\end{aligned}
$$

Observed Table

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- How to write Y_i^{obs} ?

Observed Table

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- How to write Y_i^{obs} ?

$$Y_i^{obs} = \sum_{j=1}^{J} \mathbf{1}(W_i = j) Y_i(W_i = j)$$

Orthogonal decomposition of the observed outcomes

- The potential outcome Y_i^{obs} can be written as :

$$Y_i^{obs} = \overline{Y}^{obs} + [\overline{Y}_j^{obs} - \overline{Y}^{obs}] + [Y_i^{obs} - \overline{Y}_j^{obs}]$$

where

  - $\overline{Y}_j^{obs} = \frac{1}{N_j} \sum_{i:W_i=j} Y_i^{obs}$
  - $\overline{Y}^{obs} = \sum_{j=1}^{J} \frac{N_j}{N} \overline{Y}_j^{obs}$

Orthogonal decomposition of the observed outcomes

- A measure of overall variability among the observed outcomes is the Total Sum of Squares ($SSTotal$) :

$$SSTotal = \sum_{i=1}^{N}[Y_i^{obs} - \overline{Y}^{obs}]^2 = \sum_{j=1}^{J}\sum_{i:W_i=j}[Y_i^{obs} - \overline{Y}^{obs}]^2$$

$$= SSTreatment + SSResidual$$

$$\begin{aligned} SSTreatment &= \sum_{i=1}^{N}[\overline{Y_j}^{obs} - \overline{Y}^{obs}]^2 \\ &= \sum_{j=1}^{J}\sum_{i:W_i=j}[\overline{Y_j}^{obs} - \overline{Y}^{obs}]^2 \\ &= \sum_{j=1}^{J} N_j[(\overline{Y_j}^{obs} - \overline{Y}^{obs})]^2 \end{aligned}$$

$$\begin{aligned} SSResidual &= \sum_{i=1}^{N}[Y_i^{obs} - \overline{Y_j}^{obs}]^2 \\ &= \sum_{j=1}^{J}\sum_{i:W_i=j}[Y_i^{obs} - \overline{Y_j}^{obs}]^2 \end{aligned}$$

Fisherian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Sharp null hypothesis

Fisherian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Test statistic

Fisherian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Test statistic

Source	Df	Sum of squares	Mean Squares
Treatments	J-1	SSTre	MSTre
Residuals	N-J	SSRes	MSE
Total	N-1	SSTot	

- $F = \frac{MSTre}{MSRes}$, where MSTre$=\frac{SSTre}{J-1}$ and MSRes$=\frac{SSRes}{N-J}$.

Fisherian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Test statistic
  - $\hat{\beta}$

Neymanian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Estimator for $\tau(j, j')$:

Neymanian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Estimator for $\tau(j, j')$:
  - $\hat{\tau}(j, j') = \overline{Y}_j^{obs} - \overline{Y}_{j'}^{obs}$

Neymanian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Estimator for $\tau(j, j')$:
  - $\hat{\tau}(j, j') = \overline{Y}_j^{obs} - \overline{Y}_{j'}^{obs}$
- Sampling variance of $\hat{\tau}(j, j')$:

Neymanian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Estimator for $\tau(j, j')$:
  - $\hat{\tau}(j, j') = \overline{Y}_j^{obs} - \overline{Y}_{j'}^{obs}$
- Sampling variance of $\hat{\tau}(j, j')$:
  - $Var(\hat{\tau}(j, j')) = \frac{S_j^2}{N_j} + \frac{S_{j'}^2}{N_{j'}} - \frac{S_{jj'}^2}{N_j + N_{j'}}$

Neymanian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Estimator for $\tau(j, j')$:
  - $\hat{\tau}(j, j') = \overline{Y}_j^{obs} - \overline{Y}_{j'}^{obs}$
- Sampling variance of $\hat{\tau}(j, j')$:
  - $Var(\hat{\tau}(j, j')) = \frac{S_j^2}{N_j} + \frac{S_{j'}^2}{N_{j'}} - \frac{S_{jj'}^2}{N_j + N_{j'}}$
- Estimator for the sampling variance of $\hat{\tau}(j, j')$:

Neymanian inference

Unit i	W_i	$Y_i(W_i = 1)$	$Y_i(W_i = 2)$	...	$Y_i(W_i = J)$
1	2	?	$Y_1(W_1 = 2)$	...	?
2	1	$Y_2(W_2 = 1)$	?	...	?
...	...	...	...	...	...
...	...	...	...	...	...
N	J	?	?	...	$Y_N(W_N = J)$

- Estimator for $\tau(j, j')$:
  - $\hat{\tau}(j, j') = \overline{Y}_j^{obs} - \overline{Y}_{j'}^{obs}$
- Sampling variance of $\hat{\tau}(j, j')$:
  - $Var(\hat{\tau}(j, j')) = \frac{S_j^2}{N_j} + \frac{S_{j'}^2}{N_{j'}} - \frac{S_{jj'}^2}{N_j + N_{j'}}$
- Estimator for the sampling variance of $\hat{\tau}(j, j')$:
  - $\hat{Var}(\hat{\tau}(j, j')) = \frac{s_j^2}{N_j} + \frac{s_{j'}^2}{N_{j'}}$ [Assuming constant and additive $\tau_i(j, j')$]